

A Survey of Dynamic Spectrum Access: SP & Networking Perspectives



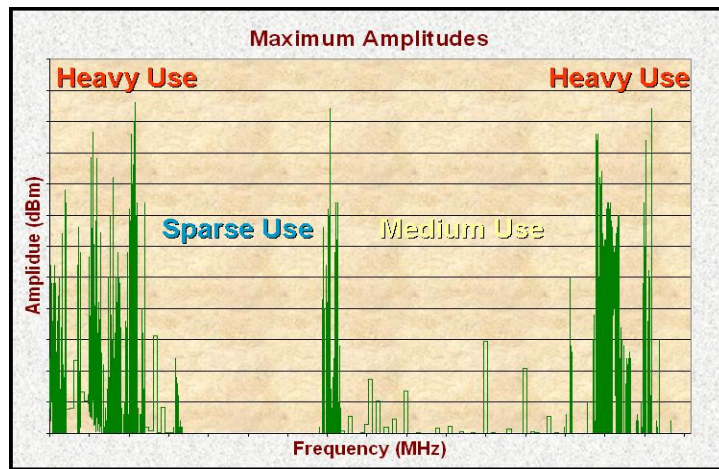
Qing Zhao and Ananthram Swami

- ▣ A Taxonomy of DSA
- ▣ Technical Challenges
- ▣ Recent Developments

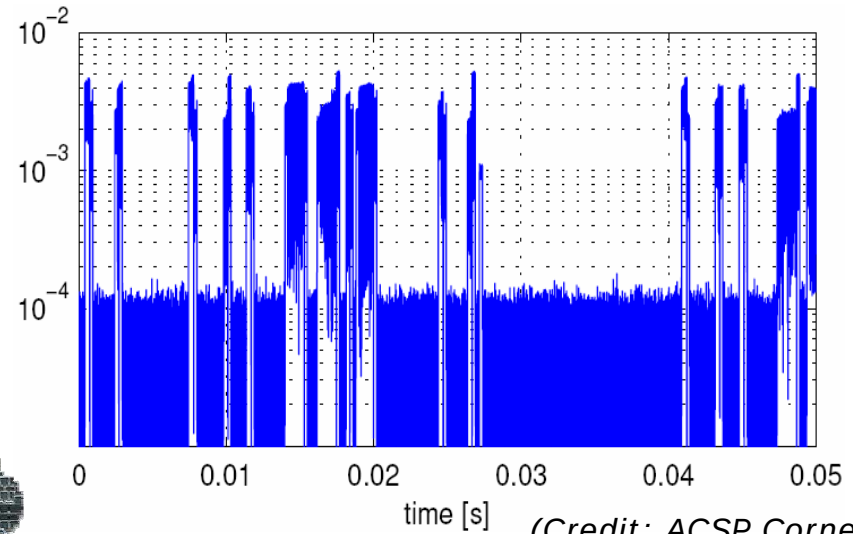
Q. Zhao, B.M. Sadler, "A Survey of Dynamic Spectrum Access: Signal Processing, Networking, and Regulatory Policy," IEEE Signal Processing Magazine, May, 2007.

- Fixed allocation
- Little Sharing
- Rigid requirements on how to use

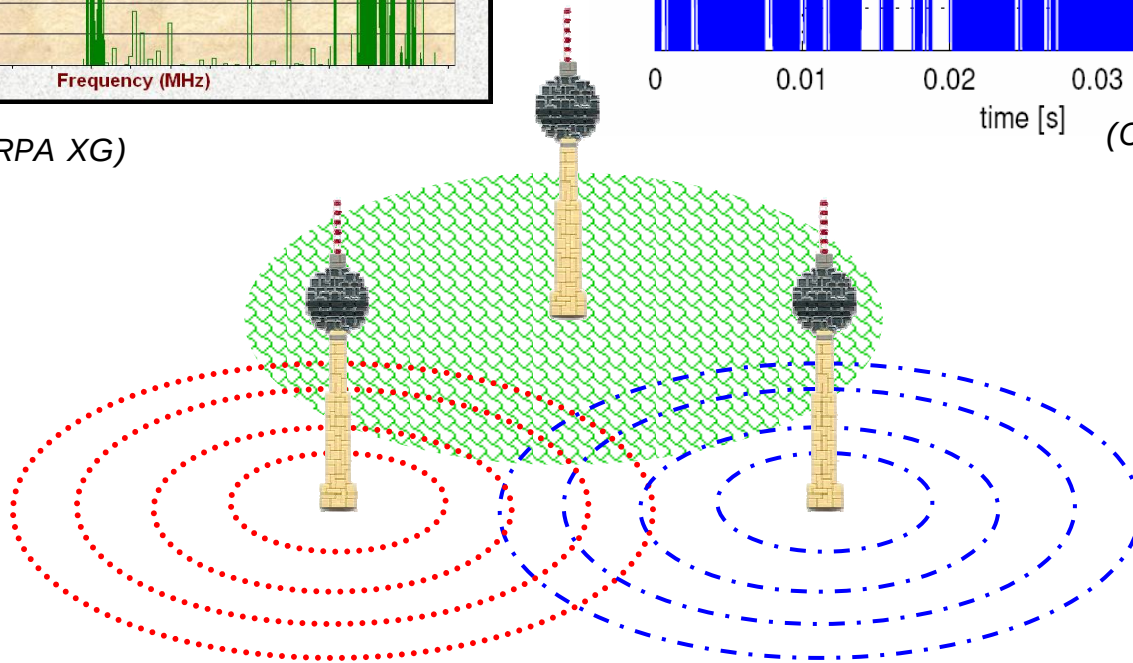
Spectrum Opportunities in Space, Time, & Frequency



(Credit: DARPA XG)



(Credit: ACSP Cornell)



Diverse Ideas, Confusing Terms

- ❑ Dynamic spectrum access
- ❑ Dynamic spectrum allocation
- ❑ Spectrum property rights
- ❑ Spectrum commons
- ❑ Opportunistic spectrum access
- ❑ Spectrum pooling
- ❑ Spectrum underlay
- ❑ Spectrum overlay
- ❑ Cognitive radio
- ❑ ...
- ❑ ...

A Taxonomy of DSA

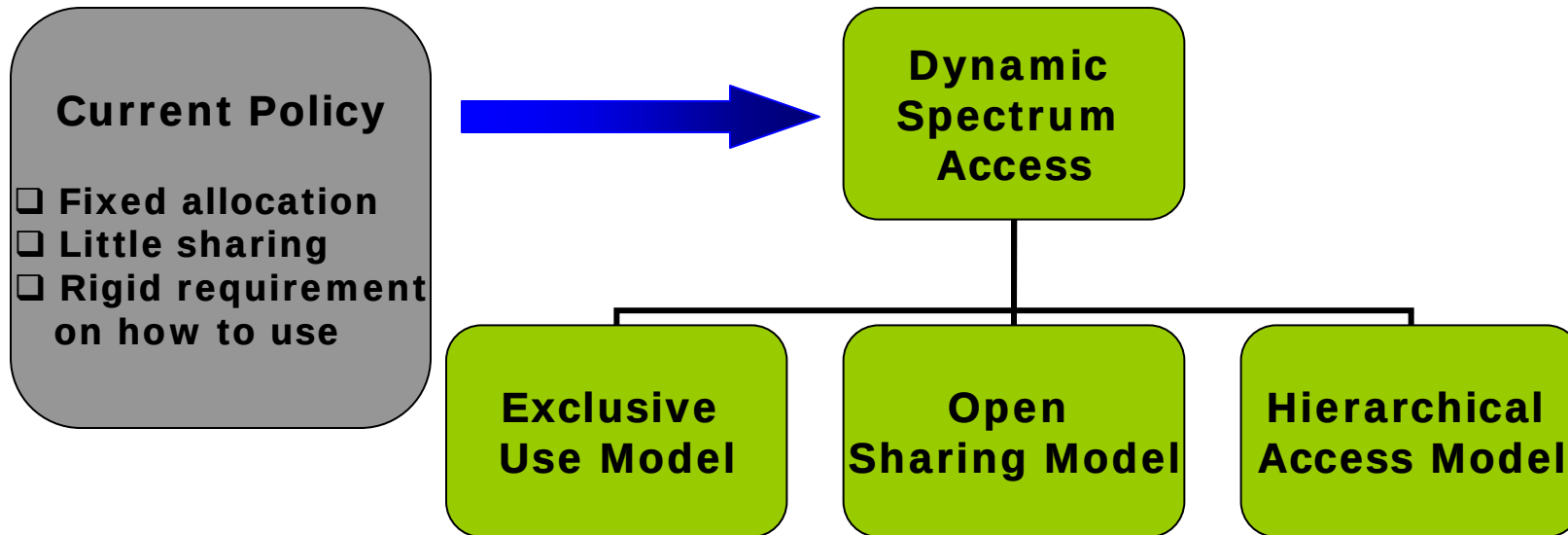
Current Policy

- ☐ Fixed allocation
- ☐ Little sharing
- ☐ Rigid requirement on how to use

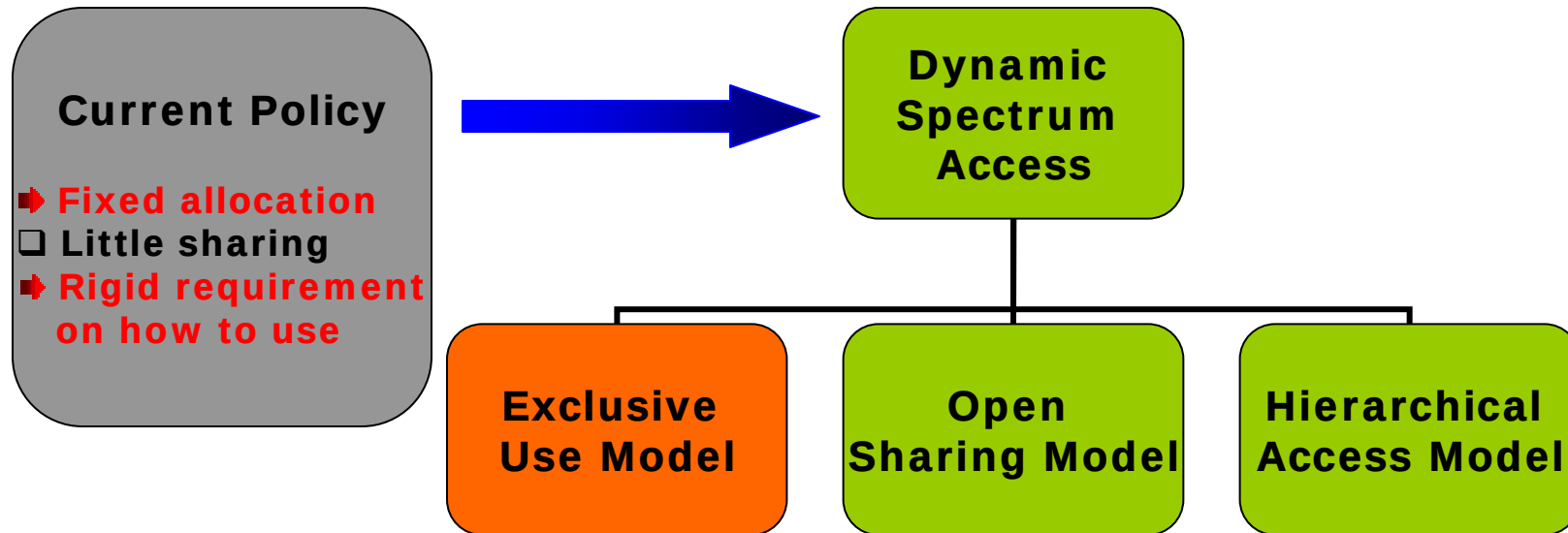


Dynamic Spectrum Access

Three DSA Models

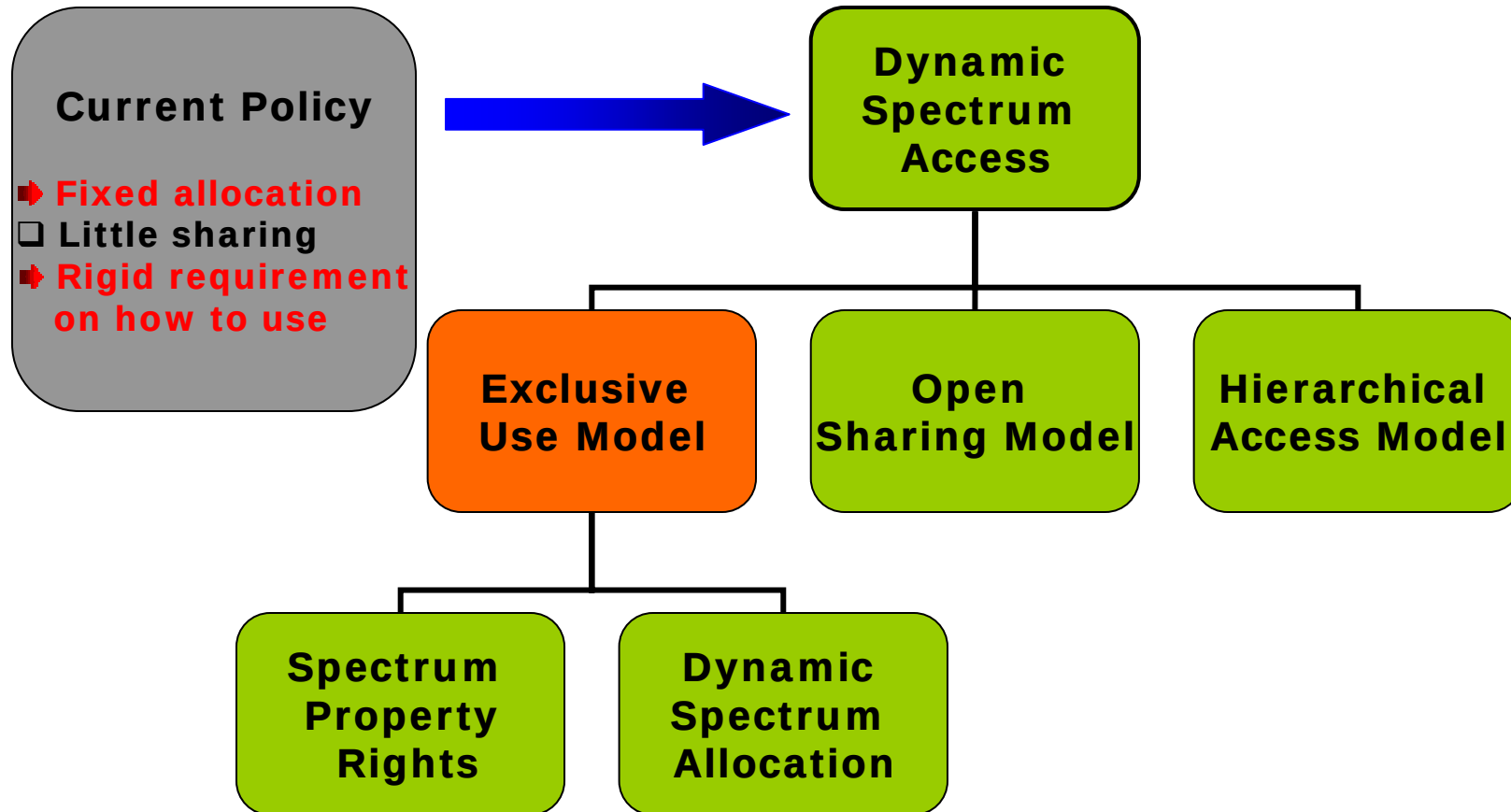


Exclusive Use Model



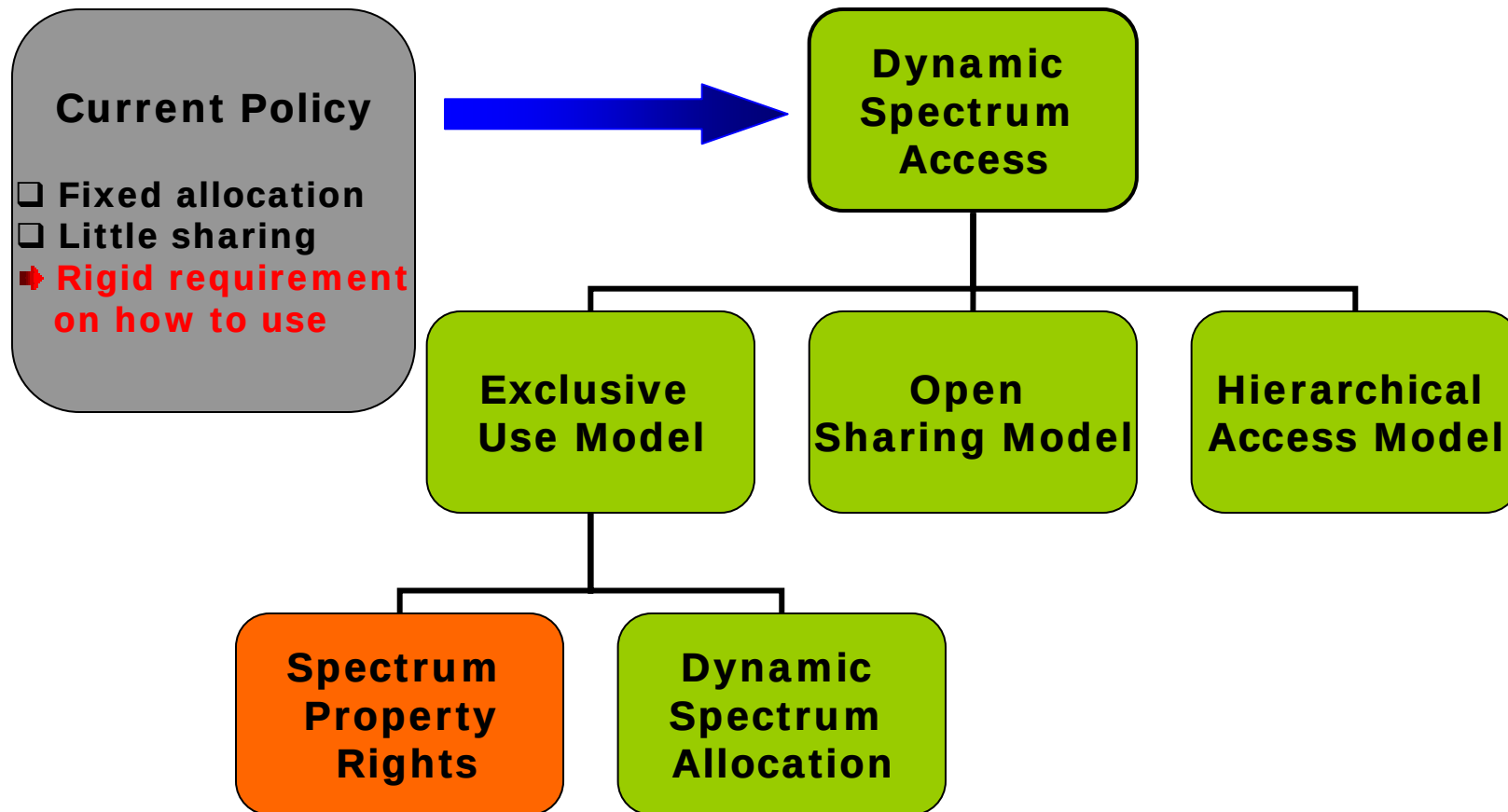
- ❑ Maintains the basic structure: license for exclusive use
- ❑ Introduces flexibility in allocation and spectrum usage

Exclusive Use Model



- ☐ Maintains the basic structure: license for exclusive use
- ☐ Introduces flexibility in allocation and spectrum usage

Spectrum Property Rights



- ❑ Allows selling and trading spectrum and freely choosing technology
- ❑ Let economy & market determine the most profitable use of spectrum

Spectrum Property Rights

How To Define and Enforce?

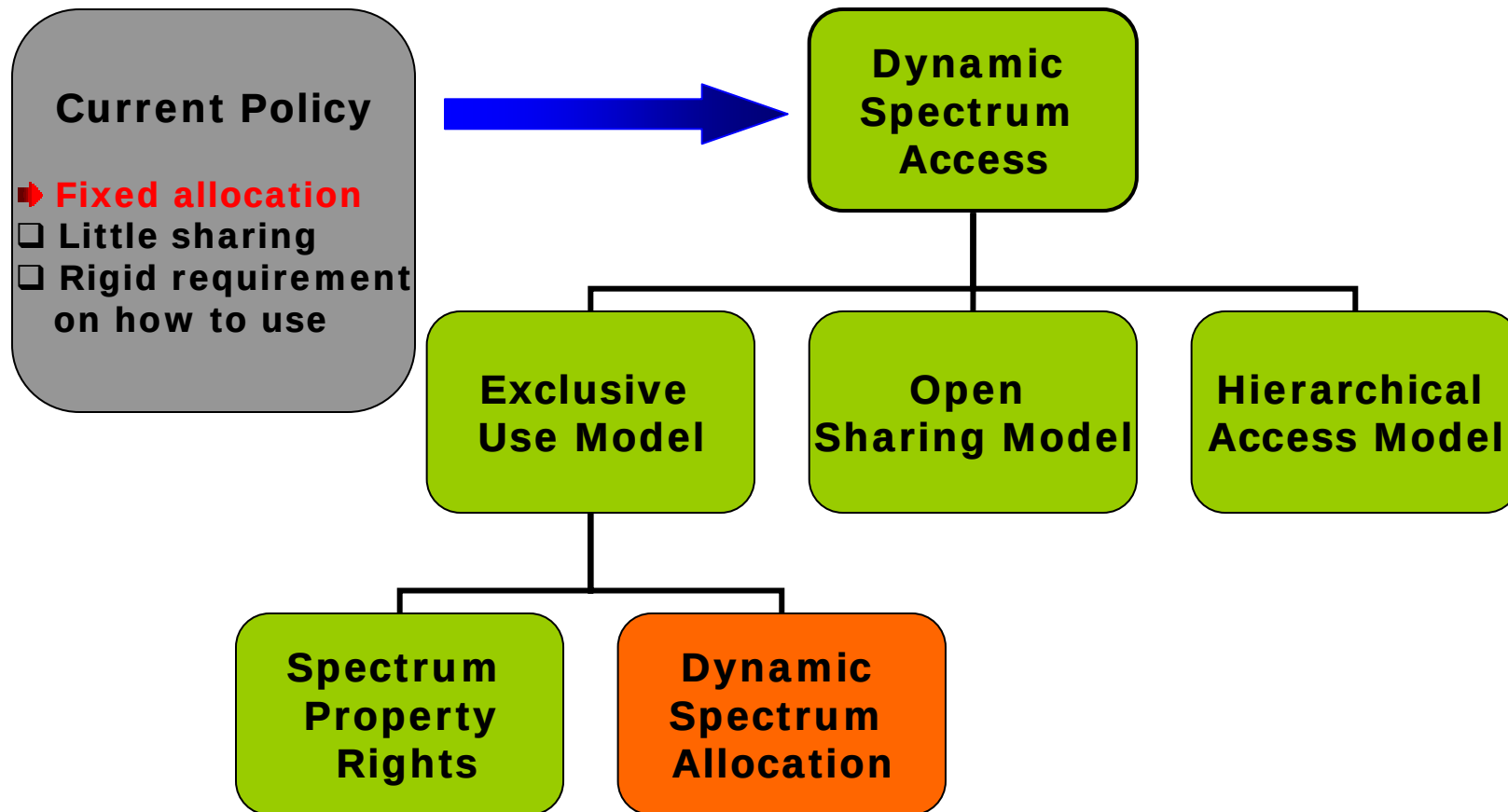


- ❑ Spatial Spillover
 - Measured or computed?
- ❑ Spectral Spillover
 - Tx or Rx's responsibility?
- ❑ How to detect trespassing

(D. Hatfield and P. Weiser, 2005 DySPAN)

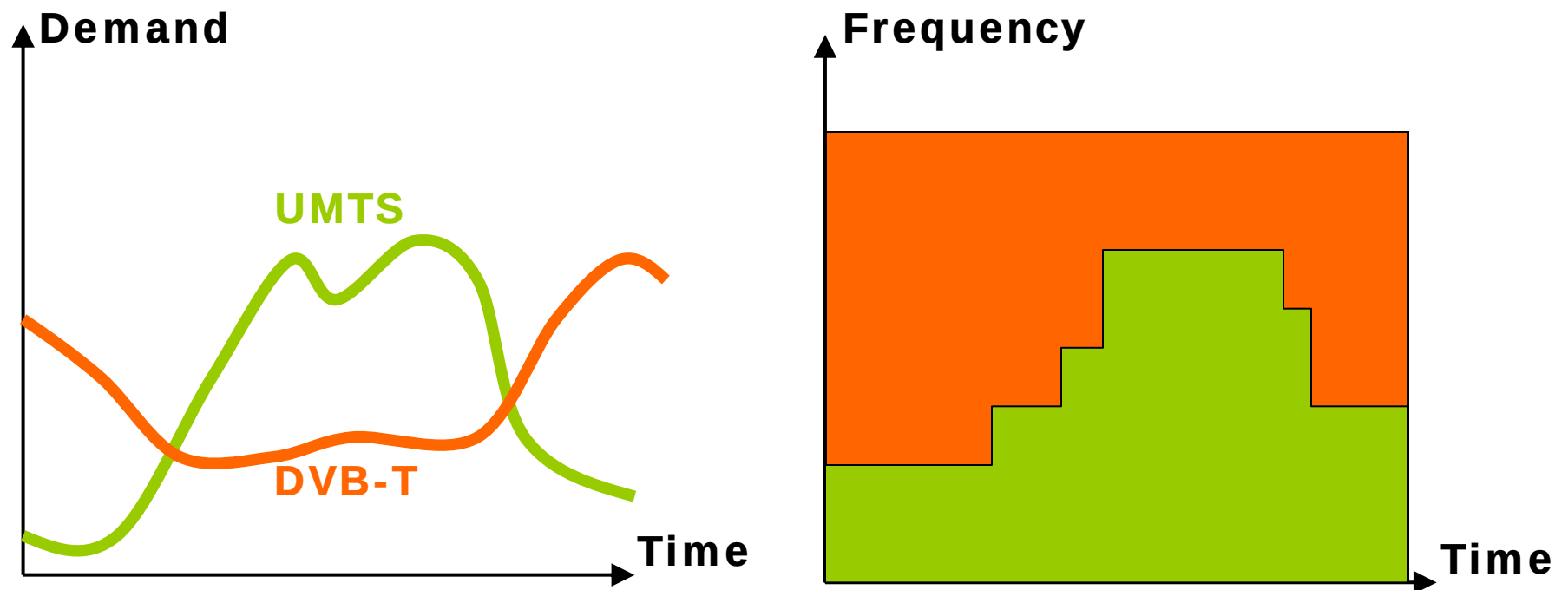
***Need Accurate Yet Simple Signal Propagation Models
And Filter Design For Effective Spillover Suppression.***

Dynamic Spectrum Allocation



- ❑ Dynamic spectrum assignment to different services
- ❑ Exploiting spatial and temporal traffic statistics

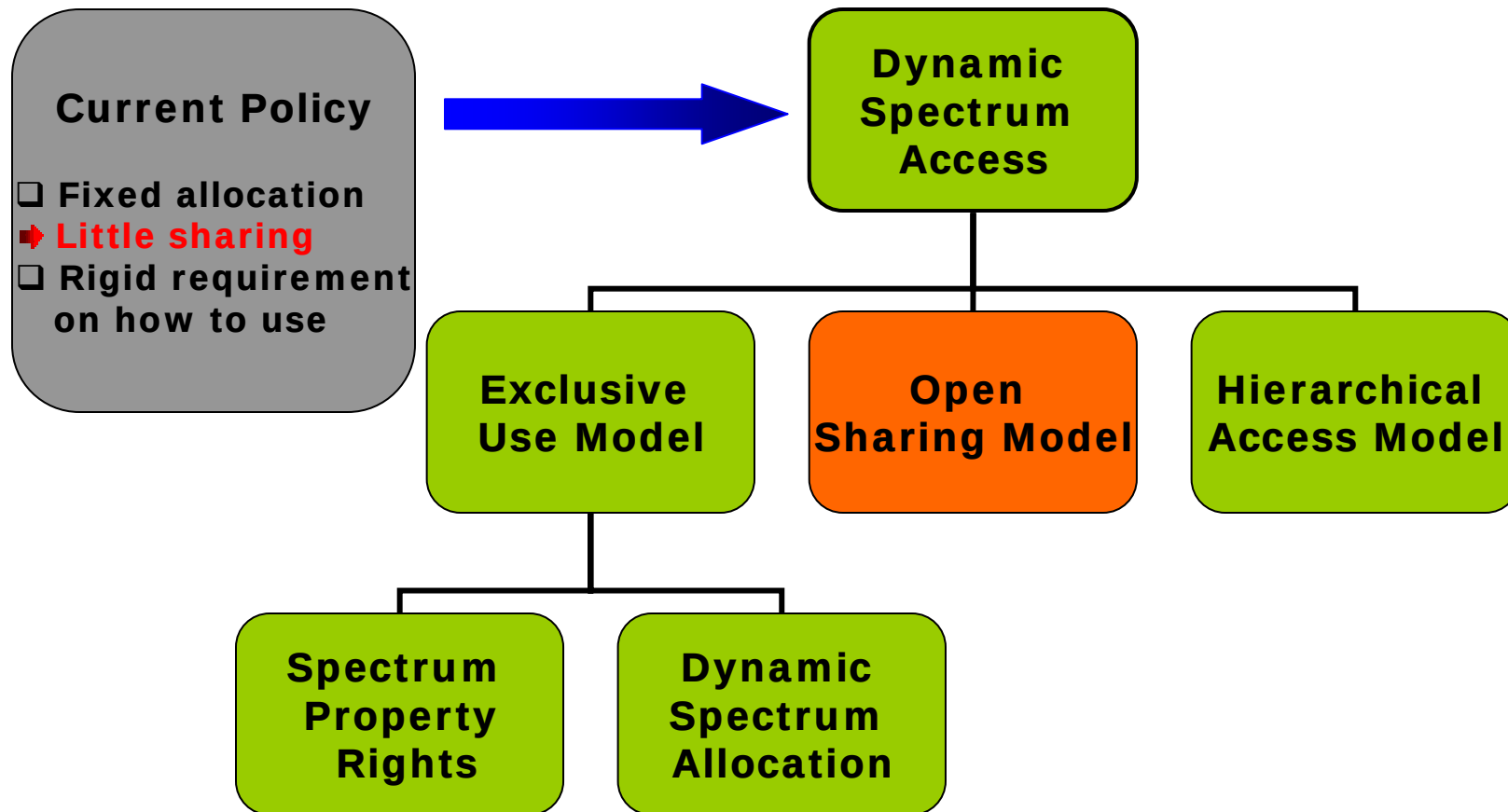
Dynamic Spectrum Allocation



- Traffic statistical modeling, estimation, and prediction
- Centralized and distributed spectrum allocation

(L. Xu, et al, 2000, P. Leves, et al, 2004)

Open Sharing Model



- ❑ Open sharing among peer users (spectrum commons)
- ❑ Draws support from the success of unlicensed ISM bands

Open Sharing Model

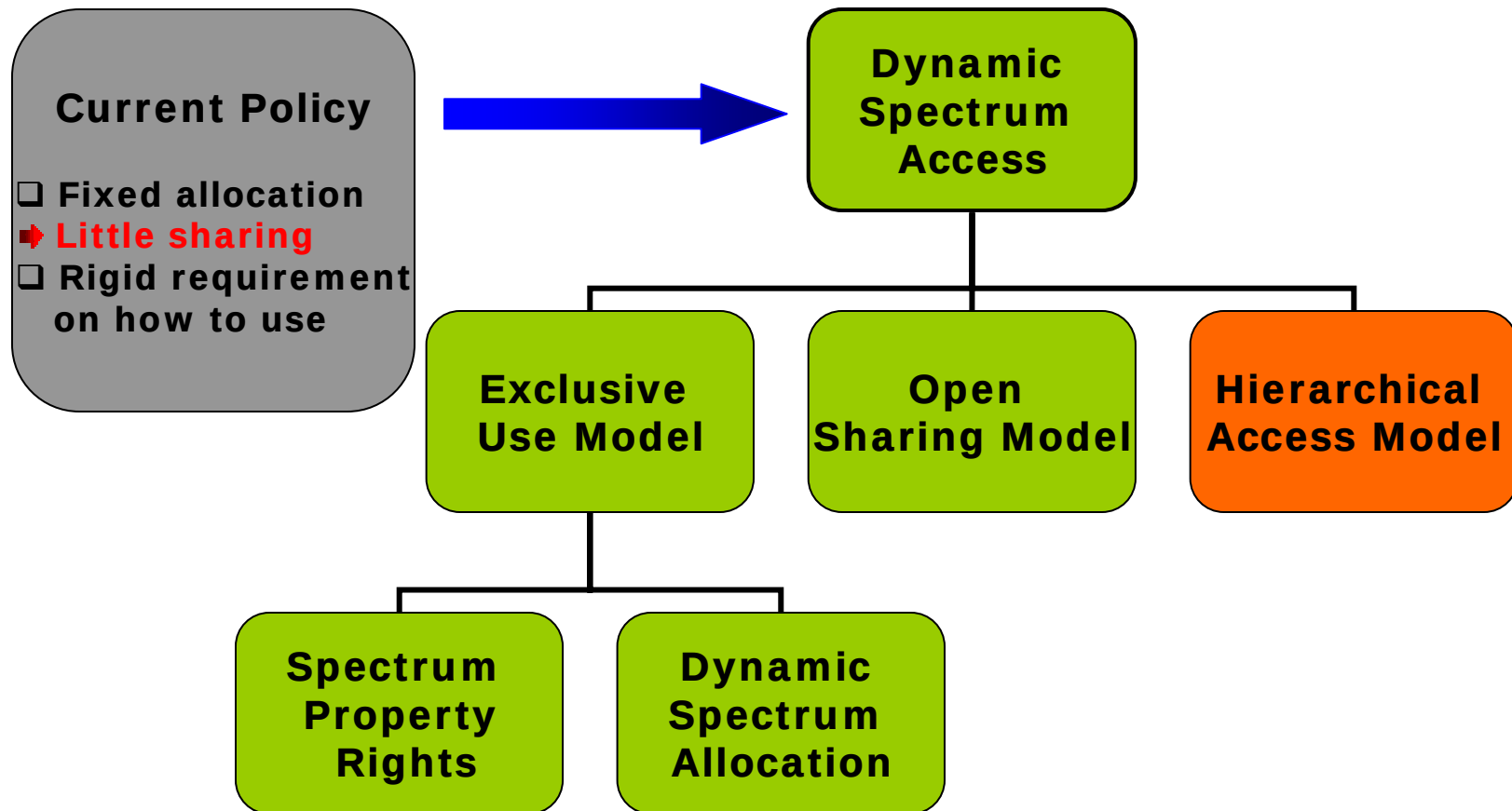
■ **Sharing among peer users**

- Users may use different services and standards
- Centralized spectrum sharing
(C. Raman, etal, 05, V. Brik, etal, 05, J. Huang, etal, 04.)
- Distributed sharing and power control
(S. Chung, etal, 03, R. Etkin, etal, 05, J. Huang, etal, 05, N. Clemens & C. Rose, 05.)

■ **Incorporating economic models**

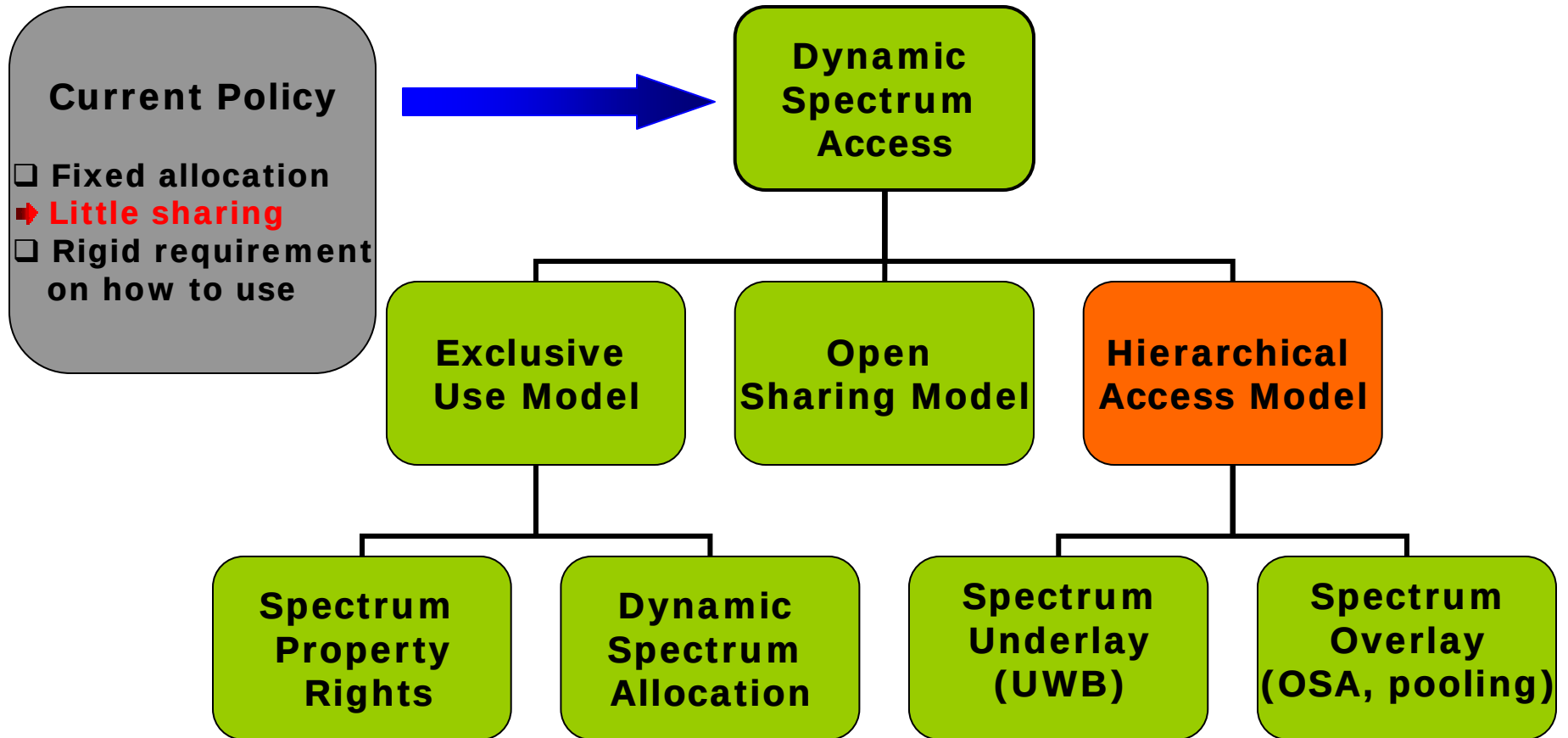
- Auction based, pricing schemes, game theory
(O. Ileri, etal, 05, S. Chung, etal, 03, R. Etkin, etal, 05, J. Huang, etal, 04, N. Clemens & C. Rose, 05.)

Hierarchical Access Model



- ❑ Hierarchical access with primary and secondary users
- ❑ sharing with limited interference to primary users (licensees)

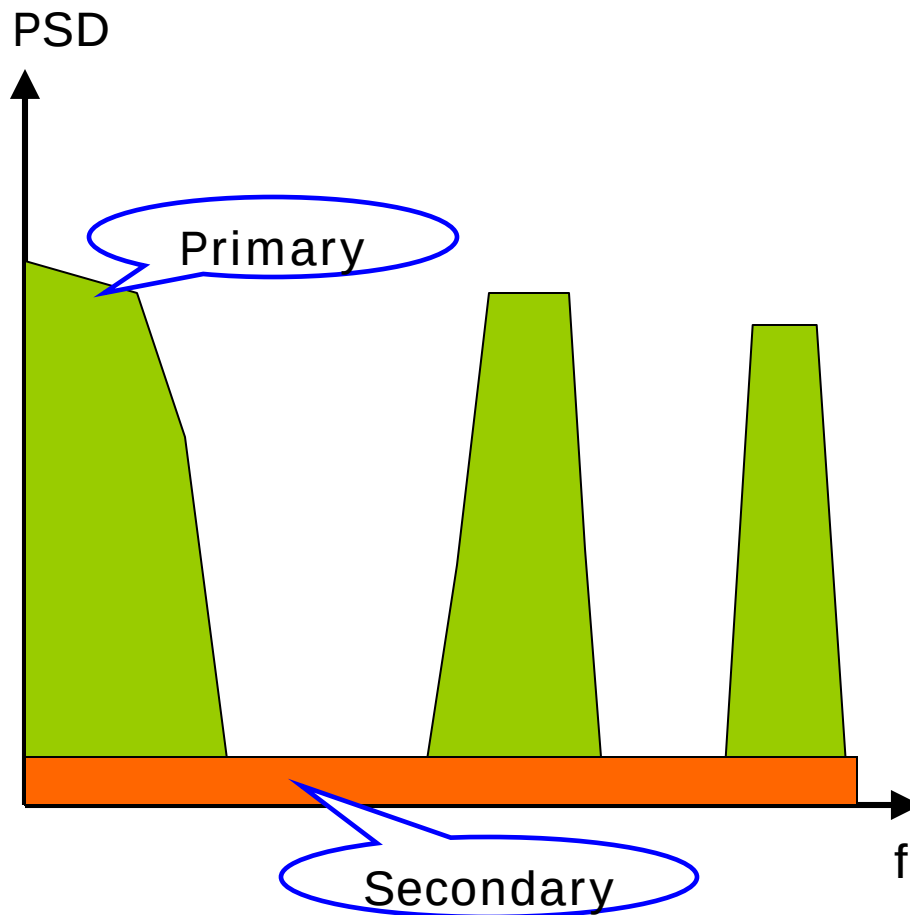
Hierarchical Access Model



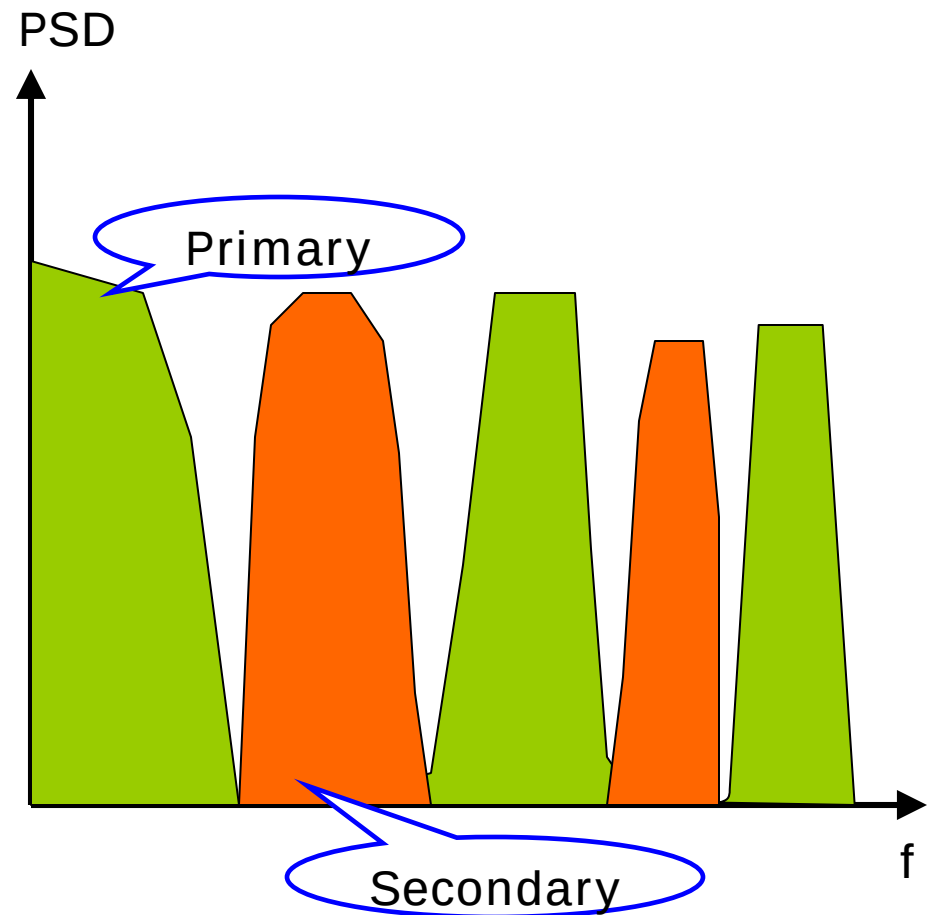
- ☐ Spectrum underlay: constraint on transmission power
- ☐ Spectrum overlay: constraint on when and where to transmit

Underlay vs. Overlay

Spectrum Underlay (UWB)



Spectrum Overlay (OSA)



Basic Concepts in OSA

□ Spectrum Opportunity

Is listen-before-talk enough?

Detecting primary transmitters or receivers?

□ Interference Constraint

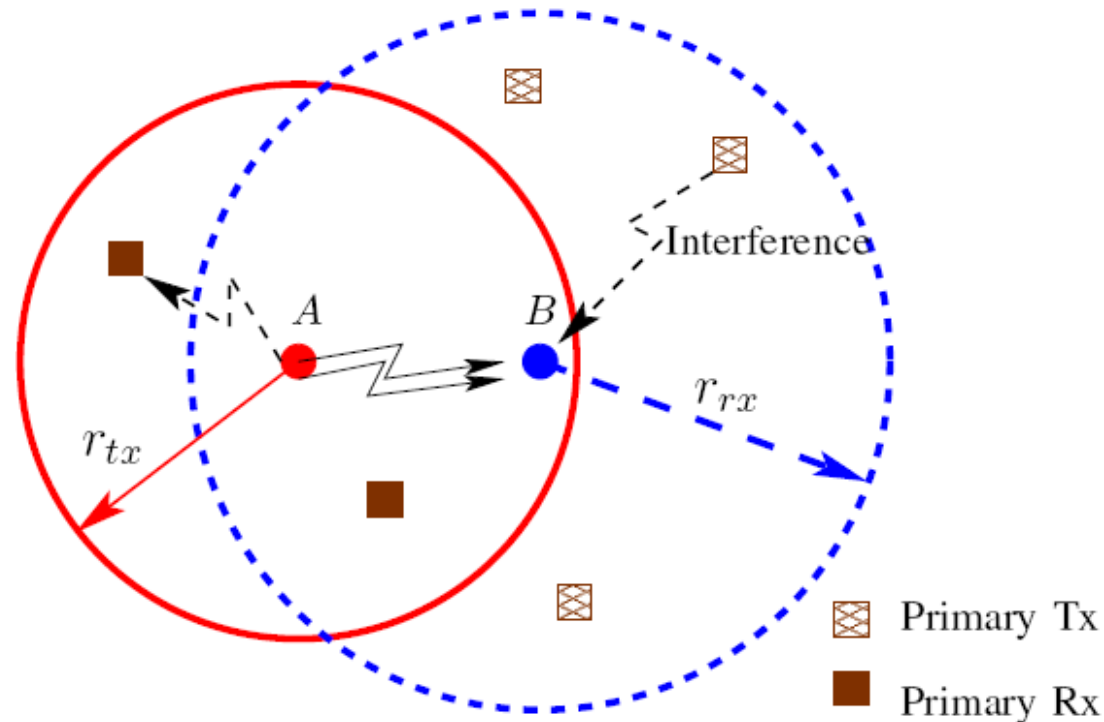
What are the minimal set of parameters?

How does it translate to the level of protection?

Aggregated interference to node-level constraint?

(Q. Zhao, ICASSP 2007.)

Spectrum Opportunity



- ❑ Channel is available at A (tx) if no primary rx **nearby**.
- ❑ Channel is available at B (rx) if no primary tx **nearby**.
- ❑ Channel is an opp. if available at both A and B.

Interference Constraint

Minimal Set of Parameters:

- Max interference power level **P** at primary rx
 - Specifies the noise floor of primary users
 - Inherent to the definition of opportunity
 - Determines tx power of secondary users

- Max probability of exceeding **P**
 - Positive to accommodate sensing errors
 - Determines spectrum access decisions

(Q. Zhao, ICASSP 2007.)

Technical Challenges in OSA

□ **Spectrum Opportunity Identification**

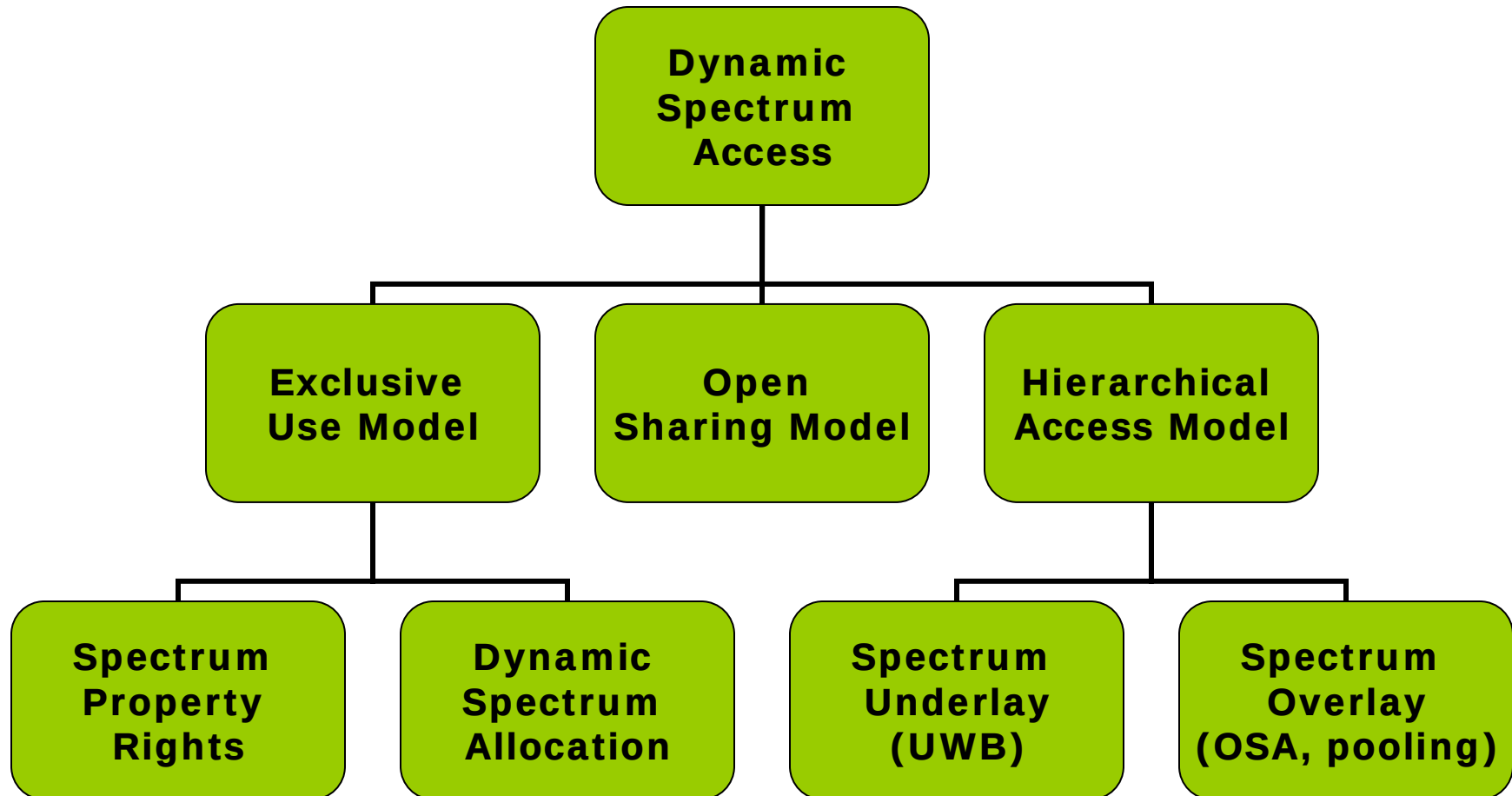
- Opportunity detection
- Opportunity tracking
- Collaborative sensing
- Overhead and time scale

□ **Spectrum Opportunity Exploitation**

- Whether to access given sensing errors occur
- How to access: modulation, tx power
- Opportunity sharing among secondary users

(Q. Zhao, A. Swami, "A Decision Theoretic Framework for Opportunistic Spectrum Access," IEEE Wireless Comm. Magazine, June, 2007.)

A Taxonomy of DSA



Cognitive Radio

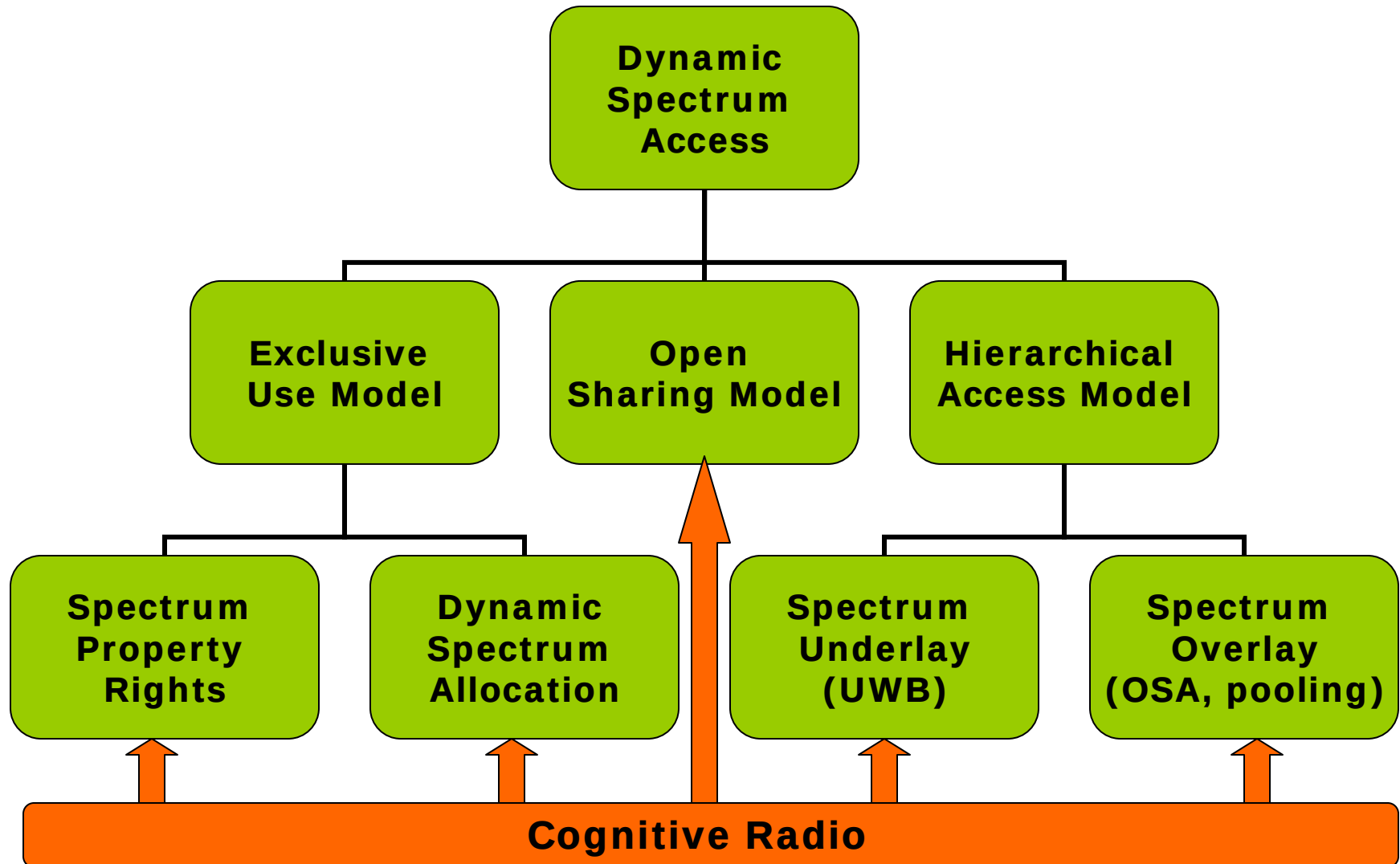
❑ Software Defined Radio

- Promoted by Mitola in 1991
- A multiband radio supporting multiple air interfaces and reconfigurable through software

❑ Cognitive Radio

- Promoted by Mitola in 1998
- Built upon a software defined radio platform
- Context-aware, autonomous reconfigurable
- Learning from and adapting to environment
- Applications ***not limited to*** dynamic spectrum access

Cognitive Radio: The Physical Platform



Recent Developments

- ❑ JSAC Special Issue, April, 2007.
- ❑ Communications Magazine, May, 2007
- ❑ Wireless Comm. Magazine, June, 2007
- ❑ JSAC, 2008.
- ❑ JSTSP, 2008.
- ❑ DySPAN 2005, 2007
- ❑ CrownCom 2006, 2007
- ❑ CogNet 2007, CWNNet 2007
- ❑ ...

Session Outline

- ❑ On the Impact of Reconfigurable Antenna Arrays in Cognitive Radio: *A. Sayeed & V. Raghavan*
- ❑ Compressed Sensing for Wideband Cognitive Radios: *Z. Tian & G.B. Giannakis*
- ❑ Squeezing the Most out of Cognitive radio: A Joint PHY/MAC Perspective: *S. Nandagopalan*
- ❑ Coordination of Allocation Measurements in OFDM Based Ad Hoc Overlay Systems: *U. Berthold, H. Heimpel, F. Jondral*
- ❑ Cognitive Dynamic Systems: *S. Haykin*